

Symbols

Sets

Name	Domains	Description
i	*	
ip	*	
k	*	

Parameters

Name	Domains	Description
P	i	
Risk	i	
A	i,ip	
B	i,k	
CapSub	k	

Variables

Name	Domains	Description
Y	i	
x	i,ip	
w	i,k	

Equations

Name	Domains	Description
demand_cov	i	
reach_shelter	i,ip	
reach_subway	i,k	
cap_shelter	ip	
cap_subway	k	
target_risk_rule	i	

Model Definition

min $\sum_i Y_i$
s.t.

demand_cov_i

$$\sum_{ip} x_{i,ip} + \sum_k w_{i,k} = P_i \quad \forall i$$

reach_shelter_{i,ip}

$$x_{i,ip} \leq P_i \cdot A_{i,ip} \quad \forall i, ip$$

reach_subway_{i,k}

$$w_{i,k} \leq P_i \cdot B_{i,k} \quad \forall i, k$$

cap_shelter_{ip}

$$\sum_i x_{i,ip} \leq 20000 \cdot Y_{ip} \quad \forall ip$$

cap_subway_k

$$\sum_i w_{i,k} \leq CapSub_k \quad \forall k$$

target_risk_rule_i

$$Y_i \cdot 20000 \geq 0.25 \cdot Risk_i \cdot P_i \quad \forall i$$

$$Y \in \mathbb{Z}_+ \quad \forall i$$

$$x \geq 0 \quad \forall i, ip$$

$$w \geq 0 \quad \forall i, k$$